

$$\nabla \cdot (\vec{g} \times \vec{f}) = \vec{f} \cdot (\nabla \times \vec{g}) - \vec{g} \cdot (\nabla \times \vec{f})$$

$$\nabla \times (\vec{g} \times \vec{f}) = (\vec{f} \cdot \nabla) \vec{g} + (\nabla \cdot \vec{f}) \vec{g} - (\vec{g} \cdot \nabla) \vec{f} - (\nabla \cdot \vec{g}) \vec{f}$$

$$\nabla (\vec{g} \cdot \vec{f}) = \vec{f} \times (\nabla \times \vec{g}) + (\vec{f} \cdot \nabla) \vec{g} + \vec{g} \times (\nabla \times \vec{f}) + (\vec{g} \cdot \nabla) \vec{f}$$

$$\left\{ \begin{array}{l} \vec{D} = \epsilon \vec{E} \\ \vec{B} = \mu \vec{H} \\ \vec{j} = \sigma \vec{E} \end{array} \right. \quad \left\{ \begin{array}{l} \frac{\rho_p}{D} = -\frac{\nabla \cdot \vec{P}}{\epsilon_0 \vec{E} + \vec{P}} \\ \vec{J}_p = \frac{\partial \vec{P}}{\partial t} \\ \vec{J}_m = \nabla \times \vec{M} \\ \vec{H} = \frac{\vec{E}}{\mu_0} - \vec{M} \end{array} \right.$$

$$\nabla f(u) = \frac{df}{du} \nabla u$$

$$\nabla \cdot \vec{A}(u) = \nabla u \cdot \frac{d\vec{A}}{du}$$

$$\nabla \times \vec{A}(u) = \nabla u \times \frac{d\vec{A}}{du}$$

$$\nabla \frac{1}{r} = -\nabla' \frac{1}{r} = -\frac{\vec{r}}{r^3}$$

$$\nabla \cdot \vec{r} = 3$$

$$\nabla \times \vec{r} = 0$$

库仑定律

$$\vec{F} = q\vec{E}$$

$$\vec{E} = \sum_i \frac{Q_i \vec{r}_i}{4\pi\epsilon_0 r_i^3}$$

毕奥-萨伐尔定律

$$\frac{d\vec{F}}{dV} = Id\vec{l} \times \vec{B}$$

$$\vec{B}(\vec{x}) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{x}') \times \vec{r}}{r^3} dV'$$

$$= \frac{\mu_0}{4\pi} \oint_L \frac{Id\vec{l} \times \vec{r}}{r^3}$$

能量密度

$$w = \frac{1}{2} (\vec{E} \cdot \vec{D} + \vec{H} \cdot \vec{B})$$

能流密度

$$\vec{S} = \vec{E} \times \vec{H}$$

洛伦兹力密度

$$\vec{f} = \rho \vec{E} + \vec{j} \times \vec{B}$$

$$\frac{dU_{\text{总}}}{dt} = \int_V \vec{f} \cdot \vec{v} = -\oint_S \vec{S} \cdot d\vec{S} - \frac{d}{dt} \int_V w d\tau$$

$$p = \vec{j} \cdot \vec{E} = \vec{f} \cdot \vec{v}$$

$$b = R_1^2 / a$$

$$Q' = -R_1 Q / a$$

$$\varphi = \frac{1}{4\pi\epsilon_0} \left[\frac{Q}{\sqrt{R^2 + a^2 - 2Ra\cos\theta}} - \frac{R_1 Q / a}{\sqrt{R^2 + R_1^4 / a^2 - 2R_1^2 R\cos\theta / a}} \right]$$

$$\nabla \cdot \frac{\vec{r}}{r^3} = \begin{cases} 0 & (r \neq 0) \\ 4\pi & (r = 0) \end{cases}$$

$$\nabla \times \frac{\vec{r}}{r^3} = \nabla \times (-\nabla \frac{1}{r}) = -\nabla \times (\nabla \frac{1}{r}) = 0$$

$$(\vec{a} \cdot \nabla) \vec{r} = \vec{a} \cdot \nabla \vec{r} = \vec{a} \cdot \vec{I} = \vec{a}$$

柱坐标系

$$\nabla \varphi = \vec{e}_r \frac{\partial \varphi}{\partial r} + \vec{e}_\phi \frac{1}{r} \frac{\partial \varphi}{\partial \phi} + \vec{e}_z \frac{\partial \varphi}{\partial z}$$

$$\nabla \cdot \vec{A} = \frac{1}{r} \frac{\partial}{\partial r} (r A_r) + \frac{1}{r} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

$$\nabla \times \vec{A} = \left(\frac{1}{r} \frac{\partial A_z}{\partial \phi} - \frac{\partial A_\phi}{\partial z} \right) \vec{e}_r + \left(\frac{\partial A_r}{\partial z} - \frac{\partial A_z}{\partial r} \right) \vec{e}_\phi + \left[\frac{1}{r} \frac{\partial}{\partial r} (r A_\phi) - \frac{1}{r} \frac{\partial A_r}{\partial \phi} \right] \vec{e}_z$$

$$\nabla^2 \varphi = \frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \varphi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial^2 \varphi}{\partial \theta^2} + \frac{\partial^2 \varphi}{\partial z^2}$$

球坐标系

$$\nabla \varphi = \vec{e}_r \frac{\partial \varphi}{\partial r} + \vec{e}_\theta \frac{1}{r} \frac{\partial \varphi}{\partial \theta} + \vec{e}_\phi \frac{1}{r \sin \theta} \frac{\partial \varphi}{\partial \phi}$$

$$\nabla \cdot \vec{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta A_\theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

$$\nabla \times \vec{A} = \frac{1}{r \sin \theta} \left[\frac{\partial}{\partial \theta} (\sin \theta A_\phi) - \frac{\partial A_\phi}{\partial \phi} \right] \vec{e}_r$$

$$+ \frac{1}{r} \left[\frac{1}{\sin \theta} \frac{\partial A_r}{\partial \phi} - \frac{\partial}{\partial r} (r A_\phi) \right] \vec{e}_\theta + \frac{1}{r} \left[\frac{\partial}{\partial r} (r A_\theta) - \frac{\partial A_r}{\partial \theta} \right] \vec{e}_\phi$$

球坐标系中拉普拉斯方程及其解

$$\nabla^2 \varphi = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \varphi}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial \varphi}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 \varphi}{\partial \phi^2} = 0$$

$$\varphi(r, \theta, \phi) = \sum_{n,m} (A_{nm} r^n + \frac{B_{nm}}{r^{n+1}}) P_n^m(\cos \theta) \cos(m\phi)$$

$$+ \sum_{n,m} (C_{nm} r^n + \frac{D_{nm}}{r^{n+1}}) P_n^m(\cos \theta) \sin(m\phi)$$

轴对称系统

$$\varphi(r, \theta) = \sum_{n=0} (A_n r^n + \frac{B_n}{r^{n+1}}) P_n(\cos \theta)$$

球对称系统

$$\varphi(r) = A + \frac{B}{r}$$

$$Q = \int_V \rho(\vec{r}') dV' \quad \vec{P} = \int_V \rho(\vec{r}') \vec{r}' dV'$$

$$\varphi^{(0)} = \frac{Q}{4\pi\epsilon_0 |\vec{r}|} \quad \varphi^{(1)} = \frac{-1}{4\pi\epsilon_0} \vec{P} \cdot \nabla \frac{1}{|\vec{r}|}$$

$$\mathcal{Q}_\theta = \int_V 3x'_i x'_j \rho(\vec{r}') dV' = \int_V 3\vec{r}' \vec{r}' \cdot \rho(\vec{r}') dV'$$

$$\varphi^{(2)} = \frac{1}{4\pi\epsilon_0} \frac{1}{6} \sum_{i,j} \mathcal{Q}_{ij} \frac{\partial^2}{\partial x_i \partial x_j} \frac{1}{|\vec{r}|}$$

约化电四极矩 $\Theta_i = \int_V (3x'_i x'_j - r'^2 \delta_{ij}) \rho(\vec{r}') dV'$

$$\vec{S}_{av} = \text{Re} \left[\frac{1}{2} \vec{E} \times \vec{H}^* \right] \quad \langle P_I \rangle = \text{Re} \left[\frac{1}{2} \vec{j}^* \cdot \vec{E} \right]$$

$$H \cos \theta - H' \cos \theta' = H'' \cos \theta'' \quad \frac{E'_{0\perp}}{E_{0\perp}} = -\frac{\sin(\theta - \theta'')}{\sin(\theta + \theta'')} = \frac{\eta_2 \cos \theta - \eta_1 \cos \theta''}{\eta_2 \cos \theta + \eta_1 \cos \theta''}$$

$$\frac{E''_{0\perp}}{E_{0\perp}} = \frac{2 \cos \theta \sin \theta''}{\sin(\theta + \theta'')} = \frac{2\eta_2 \cos \theta}{\eta_2 \cos \theta + \eta_1 \cos \theta''}$$

$$\frac{E'_{0\parallel}}{E_{0\parallel}} = \frac{\text{tg}(\theta - \theta'')}{\text{tg}(\theta + \theta'')} = \frac{\eta_1 \cos \theta - \eta_2 \cos \theta''}{\eta_1 \cos \theta + \eta_2 \cos \theta''}$$

$$\frac{E''_{0\parallel}}{E_{0\parallel}} = \frac{2 \cos \theta \sin \theta''}{\sin(\theta + \theta'') \cos(\theta - \theta'')} = \frac{2\eta_2 \cos \theta}{\eta_1 \cos \theta + \eta_2 \cos \theta''}$$

全反射

$$k_Z'' = i\kappa \quad \kappa = k_0 \sqrt{\sin^2 \theta - n_{21}^2}$$

$$\vec{E}'' = \vec{E}_0'' e^{-\kappa z} e^{i(k_x x - \omega t)}$$

$$\sin \theta'' \rightarrow \frac{k_x''}{k''} = \frac{\sin \theta}{n_{21}} = \frac{\sin \theta}{\sin \theta_c} \quad \cos \theta'' \rightarrow \frac{k_z''}{k''} = i \sqrt{\frac{\sin^2 \theta}{n_{21}^2} - 1}$$

介质

导电媒质

$$\frac{\partial \rho}{\partial t} + \frac{\sigma}{\epsilon} \rho = 0$$

$$\rho(t) = \rho_0 e^{-\frac{\sigma}{\epsilon} t}$$

自由电荷分布 (因 $\vec{j} = \sigma \vec{E}, \nabla \cdot \vec{j} + \frac{\partial \rho}{\partial t} = 0, \nabla \cdot \vec{D} = \rho$)

$$\nabla^2 \vec{E} + k^2 \vec{E} = 0 \quad \nabla^2 \vec{H} + k^2 \vec{H} = 0$$

$$\nabla \cdot \vec{E} = 0 \quad \vec{B} = -\frac{i}{\omega} \nabla \times \vec{E}$$

$$k^2 = \omega^2 \mu \epsilon' \quad \epsilon' = \epsilon + i \frac{\sigma}{\omega}$$

$$\vec{k} = \vec{\beta} + i \vec{\alpha} \quad \left\{ \begin{array}{l} \beta^2 - \alpha^2 = \omega^2 \mu \epsilon \\ \vec{\alpha} \cdot \vec{\beta} = \frac{1}{2} \omega \mu \sigma \end{array} \right. \quad \left\{ \begin{array}{l} \text{良导体 } \alpha \approx \beta \approx \sqrt{\frac{\omega \mu \sigma}{2}} \\ \frac{\sigma}{\epsilon \omega} \gg 1 \quad \delta = \frac{1}{\alpha} = \sqrt{\frac{2}{\mu \sigma \omega}} \end{array} \right.$$

$$\vec{E}(\vec{r}, t) = \vec{E}_0 e^{-\vec{\alpha} \cdot \vec{r}} e^{i(\vec{\beta} \cdot \vec{r} - \omega t)}$$

$$\left\{ \begin{array}{l} \beta = \omega \sqrt{\mu \epsilon} \left[\frac{1}{2} \left(1 + \sqrt{1 + \frac{\sigma^2}{\epsilon^2 \omega^2}} \right) \right]^{\frac{1}{2}} \quad \eta_2 = \sqrt{\frac{\mu_2}{\epsilon_2 + i \frac{\sigma}{\omega}}} \ll \eta_1 \\ \alpha = \omega \sqrt{\mu \epsilon} \left[\frac{1}{2} \left(\sqrt{1 + \frac{\sigma^2}{\epsilon^2 \omega^2}} - 1 \right) \right]^{\frac{1}{2}} \end{array} \right.$$

波导

$$\omega_{c,mn} = \frac{\pi}{\sqrt{\mu \epsilon}} \sqrt{\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2}$$

$$\mathcal{L}_{c,mn} = \frac{2\pi C}{\omega_{cmn}} = \frac{2}{\sqrt{\left(\frac{m}{a} \right)^2 + \left(\frac{n}{b} \right)^2}}$$

$$E_x = \frac{i}{k_t^2} \left(k_z \frac{\partial E_z}{\partial x} + \omega \mu \frac{\partial H_z}{\partial y} \right) \quad E_y = \frac{i}{k_t^2} \left(k_z \frac{\partial E_z}{\partial y} - \omega \mu \frac{\partial H_z}{\partial x} \right)$$

$$H_x = \frac{i}{k_t^2} \left(-\omega \mu \frac{\partial E_z}{\partial y} + k_z \frac{\partial H_z}{\partial x} \right) \quad H_y = \frac{i}{k_t^2} \left(\omega \mu \frac{\partial E_z}{\partial x} + k_z \frac{\partial H_z}{\partial y} \right)$$

定义矢势和标势

$$\vec{B} = \nabla \times \vec{A} \quad \vec{E} = -\nabla \varphi - \frac{\partial \vec{A}}{\partial x}$$

洛伦兹规范

$$\nabla \cdot \vec{A} + \frac{1}{c^2} \frac{\partial \varphi}{\partial t} = 0$$

达朗贝尔方程

$$\left\{ \begin{array}{l} \nabla^2 \varphi - \frac{1}{c^2} \frac{\partial^2 \varphi}{\partial t^2} = -\frac{\rho}{\epsilon_0} \\ \nabla^2 \vec{A} - \frac{1}{c^2} \frac{\partial^2 \vec{A}}{\partial t^2} = -\mu_0 \vec{j} \end{array} \right.$$

$$\varphi(\vec{x}, t) = \frac{1}{4\pi\epsilon_0} \int_V \frac{\rho(\vec{x}', t - \frac{r}{c})}{r} d\tau'$$

方程的解:
推迟势

$$\vec{A}(\vec{x}, t) = \frac{\mu_0}{4\pi} \int_V \frac{\vec{j}(\vec{x}', t - \frac{r}{c})}{r} d\tau'$$

$$\vec{j}(\vec{x}', t) = \vec{j}(\vec{x}') e^{-i\omega t} \quad t_0 = t - \frac{r}{c}$$

$$\vec{A}(\vec{x}, t) = \frac{\mu_0 e^{-i\omega t}}{4\pi} \int_V \frac{\vec{j}(\vec{x}') e^{ikr}}{r} dV'$$

时谐电流下得到 $\vec{B}$ 和 $\vec{E}$

$$\vec{E} = \frac{ic}{k} \nabla \times \vec{B}$$

电偶极辐射

$$r \approx R - \vec{e}_R \cdot \vec{x}$$

$$\vec{A}(\vec{x}) = \frac{\mu_0 e^{ikR}}{4\pi R} \int \vec{J}(\vec{x}') dV'$$

$$\dot{\vec{P}} = \int_V \vec{j} \cdot \vec{I} d\tau' = \int_V \vec{j} d\tau' = \sum_i q_i \vec{v}_i$$

$$\vec{B} = \frac{e^{ikR}}{4\pi\epsilon_0 c^3 R} \ddot{\vec{p}} \times \vec{e}_R = \frac{1}{4\pi\epsilon_0 c^3 R} e^{ikR} |\ddot{\vec{p}}| \sin(\theta) \vec{e}_\phi$$

$$\vec{E} = \frac{e^{ikR}}{4\pi\epsilon_0 c^2 R} (\ddot{\vec{p}} \times \vec{e}_R) \times \vec{e}_R = \frac{1}{4\pi\epsilon_0 c^2 R} e^{ikR} |\ddot{\vec{p}}| \sin(\theta) \vec{e}_\theta$$

平均能流密度

$$\langle \vec{S} \rangle = \frac{|\ddot{\vec{p}}|^2}{32\pi^2 \epsilon_0 c^3 R^2} \sin^2 \theta \hat{n}$$

辐射角分布

$$\overline{f(\theta \cdot \varphi)} = \frac{|\ddot{\vec{p}}|^2}{32\pi^2 \epsilon_0 c^3} \sin^2 \theta$$

辐射功率

$$\bar{P} = \frac{1}{4\pi\epsilon_0} \cdot \frac{|\ddot{\vec{p}}|^2}{3c^3}$$

作业2.8 分离变量法求接地导体球外放一个点电荷所激发的电势
解: 该情形是求Poisson方程的定解问题

$$\nabla^2 \varphi = -\frac{Q_1}{\varepsilon} \delta(\vec{r} - \vec{r}')$$

其中 $\vec{r}' = a\vec{e}_z$ 表示点电荷的位置。

易知边界条件有:

$$\begin{aligned}\varphi|_{R=R_0} &= 0 \\ \varphi|_{R \rightarrow \infty} &= 0 \\ \varphi &= u + v\end{aligned}$$

为了求解该定解问题, 这里将 φ 一分为二,

选取某个函数 v , 使其满足

$$\begin{cases} \nabla^2 v = -\frac{Q_1}{\varepsilon} \delta(\vec{r} - \vec{r}') \\ v|_{R \rightarrow \infty} = 0 \end{cases}$$

可以得到这个函数 v 在 $R = R_0$ 边界的取值

$$v|_{R=R_0} = v_0$$

那么需要让 u 满足

$$\begin{cases} \nabla^2 u = 0 \\ u|_{R=R_0} = -v_0 \\ u|_{R \rightarrow \infty} = 0 \end{cases}$$

即解Laplace方程即可。

这道题中 v 即空间中放置一孤立点电荷产生的电势

$$v = \frac{Q_1}{4\pi\epsilon r}$$

其中 r 是空间某点到该点电荷的距离

该对称条件下 u 的通解为

$$u = \sum_n \left(a_n R^n + \frac{b_n}{R^{n+1}} \right) P_n(\cos \theta)$$

带入边界条件:

$$\begin{aligned}u|_{R \rightarrow \infty} &= 0 \Rightarrow a_n = 0 \\ u|_{R=R_0} &= -v_0 = -\frac{Q_1}{4\pi\epsilon R}\end{aligned}$$

将其按Legendre多项式展开

$$\frac{1}{r} = \begin{cases} \frac{1}{a} \sum_{n=0}^{\infty} \left(\frac{R}{a} \right)^n P_n(\cos \theta) & (R < a) \\ \frac{1}{R} \sum_{n=0}^{\infty} \left(\frac{a}{R} \right)^n P_n(\cos \theta) & (R > a) \end{cases}$$

由于点电荷在求外 $R_0 < a$, 那么

$$\begin{aligned}\sum_n \left(\frac{b_n}{R^{n+1}} \right) P_n(\cos \theta) &= \frac{-Q_1}{4\pi\epsilon a} \sum_n \left(\frac{R}{a} \right)^n P_n(\cos \theta) \\ \Rightarrow b_n &= \frac{-Q_1 R_0}{4\pi\epsilon} \frac{R_0^{2n+1}}{a^{n+1}}\end{aligned}$$

因此叠加两者可得到

$$\varphi = u + v = \frac{Q_1}{4\pi\epsilon r} + \sum_{n=0}^{\infty} \frac{-Q_1}{4\pi\epsilon} \frac{R_0^{2n+1}}{a^{n+1} R^{n+1}} P_n(\cos \theta)$$

$$\begin{aligned}&= \frac{Q_1}{4\pi\epsilon r} - \frac{R_0 Q_1}{4\pi\epsilon a R} \sum_{n=0}^{\infty} \frac{R_0^{2n}}{a^n R^n} P_n(\cos \theta) \\ &= \frac{Q_1}{4\pi\epsilon r} - \frac{R_0 Q_1}{4\pi\epsilon a R} \sum_{n=0}^{\infty} \left(\frac{b}{R} \right)^n P_n(\cos \theta)\end{aligned}$$

其中 $b = \frac{R_0^2}{a}$, 再套用展开式($b < R_0 < R$), 得到

$$\begin{aligned}\varphi &= \frac{Q_1}{4\pi\epsilon r} - \frac{R_0 Q_1}{4\pi\epsilon a r} \\ r' &= \sqrt{R^2 + b^2 - 2Rb \cos \theta}\end{aligned}$$

与镜像法结果一致。

例题 2-13 如图 2-19 所示, 某球形电容器的内导体球半径为 b , 与之同心的外导体球壳的半径为 a , 其间填充两种理想介质, 介电常数分别为 ϵ_1

和 ϵ_2 。如果内导体球所带电荷为 q , 试分析:

- (1) 电容器内的电场分布情况;
- (2) 内导体球上的电荷具体分布情况。

解 根据结构的对称性, 虽然电容器上、下部分填充的介质不同, 但仍然可以认为其电场 E_1 、 E_2 沿径向, 而且电场在各自区域的半球面上均匀分布。

另外, 对两种不同介质所构成的分界面而言, 径向电场是分界面的切向分量; 根据电场的边界条件可知, 电场强度的切向分量必须连续, 即

$$E_1 = E_2 = E(r) e_r = E$$

通过上述电场强度可得其电位移变量的表达式如下:

$$D_1 = \epsilon_1 E_1 = \epsilon_1 E(r) e_r, \quad D_2 = \epsilon_2 E_2 = \epsilon_2 E(r) e_r$$

对球形电容器内半径为 r 的闭合球面应用介质内的 Gauss 定理, 可得

$$q = \oint_S D \cdot dS = \oint_S D e_r \cdot e_r dS = (D_1 + D_2) \cdot 2\pi r^2 = (\epsilon_1 + \epsilon_2) E \cdot 2\pi r^2$$

即电容器内的电场强度为

$$E = \frac{q}{2\pi(\epsilon_1 + \epsilon_2)r^2} e_r$$

其电位移变量的分布情况为

$$D_1 = \epsilon_1 E = \frac{q\epsilon_1}{2\pi(\epsilon_1 + \epsilon_2)r^2} e_r$$

$$D_2 = \epsilon_2 E = \frac{q\epsilon_2}{2\pi(\epsilon_1 + \epsilon_2)r^2} e_r$$

对内导体球, 其面电荷密度的大小等于电位移矢量在球面处的法向分量, 即

$$\rho_1 = e_n \cdot D_1|_{r=b} = e_r \cdot \frac{q\epsilon_1}{2\pi(\epsilon_1 + \epsilon_2)b^2} e_r = \frac{\epsilon_1}{2\pi(\epsilon_1 + \epsilon_2)b^2} q$$

$$\rho_2 = e_n \cdot D_2|_{r=b} = e_r \cdot \frac{q\epsilon_2}{2\pi(\epsilon_1 + \epsilon_2)b^2} e_r = \frac{\epsilon_2}{2\pi(\epsilon_1 + \epsilon_2)b^2} q$$

作业4.8 平面电磁波由真空倾斜入射到导电媒质表面上, 入射角为 θ_1 , 求导电介质中电磁波的相速度和衰减长度。若导电介质为金属, 结果又如何?

解: 界面处边值关系满足

$$\begin{aligned}k_x^{(0)} &= k_x = \beta_x + i\alpha_x \\ \Rightarrow \alpha_x &= 0, \quad \beta_x = k_x = k_x^{(0)} \\ \vec{\alpha} &= \alpha_x \vec{e}_x + \alpha_z \vec{e}_z = \alpha_z \vec{e}_z \\ \vec{\beta} &= \beta_x \vec{e}_x + \beta_z \vec{e}_z = k_x \vec{e}_x + \beta_z \vec{e}_z\end{aligned}$$

$\vec{\alpha}$ 只有 z 分量, $\vec{\beta}$ 有 x 和 z 两个方向分量。

$$k^2 = \beta^2 - \alpha^2 - 2i\vec{\alpha} \cdot \vec{\beta} = \omega^2 \mu (\epsilon + i\frac{\sigma}{\omega})$$

导电媒质中不能取近似, 就是要解下列方程:

$$\begin{cases} \beta_x^2 + \beta_z^2 - \alpha^2 = \omega^2 \mu \epsilon - \alpha^2 = \beta_x^2 + k_z^2 - \omega^2 \mu \epsilon \\ \alpha_x \beta_z = \frac{1}{2} \omega \mu \sigma \rightarrow \alpha_z^2 \beta_z^2 = \left(\frac{1}{2} \omega \mu \sigma \right)^2 \\ \beta_x = k_x = k_x^{(0)} = \frac{\omega}{c} \sin \theta_1 \end{cases}$$

记 $\gamma^2 = k_x^2 - \omega^2 \mu \epsilon$, 上述方程简化为

$$(\gamma^2 + \beta_z^2)(\beta_z^2) = \left(\frac{1}{2} \omega \mu \sigma \right)^2$$

用二次函数求根公式得

$$\beta_z^2 = \frac{-\gamma^2 \pm \sqrt{(\gamma^2)^2 + 4 \left(\frac{1}{2} \omega \mu \sigma \right)^2}}{2}$$

由于 $\beta_z^2 > 0$, 上式取+号。将 γ^2 写开即得

$$\beta_z^2 = \frac{1}{2} \left(\mu \epsilon \omega^2 - \frac{\omega^2}{c^2} \sin^2 \theta_1 \right) + \frac{1}{2} \left[\left(\mu \epsilon \omega^2 - \frac{\omega^2}{c^2} \sin^2 \theta_1 \right)^2 + (\mu \sigma \omega)^2 \right]^{\frac{1}{2}}$$

$$\alpha^2 = \alpha_z^2 = -\frac{1}{2} \left(\mu \epsilon \omega^2 - \frac{\omega^2}{c^2} \sin^2 \theta_1 \right) + \frac{1}{2} \left[\left(\mu \epsilon \omega^2 - \frac{\omega^2}{c^2} \sin^2 \theta_1 \right)^2 + (\mu \sigma \omega)^2 \right]^{\frac{1}{2}}$$

$$\beta_x = \frac{\omega}{c} \sin \theta_1, \quad \beta_y = 0 \quad \text{若是良导体: } \alpha_z \beta_z \gg \frac{1}{2} k^{(0)2} = \frac{1}{2} \beta_x^2$$

$$\text{相速 } v = \omega / \beta, \quad \text{衰减长度} = 1 / \alpha \Rightarrow \begin{cases} \beta_x \ll \beta_z \\ \alpha_x \approx \beta_z \approx \sqrt{\frac{\omega \mu \sigma}{2}} \end{cases}$$

例题 3-7 磁导率为 μ , 内外半径分别为 a, b 的无限长空心圆柱体, 其中存在轴向均匀电流密度 J , 求各处的磁场强度和磁化电流密度。

解 取半径为 r 的积分环路, 根据式 (3-32) 安培环路定理得

$$\oint_H H \cdot dl = \int_S J \cdot dS = I$$

其中, I 为积分环路。磁场沿 e_ϕ 方向。于是

当 $0 < r \leq a$ 时, $J = 0, I = 0$, 所以 $H = 0$;

$$\text{当 } a < r \leq b \text{ 时, } I = J(\pi r^2 - \pi a^2), \text{ 所以 } H = \frac{J(r^2 - a^2)}{2r} e_\phi;$$

$$\text{当 } r > b \text{ 时, } I = J(\pi b^2 - \pi a^2), \text{ 所以 } H = \frac{J(b^2 - a^2)}{2r} e_\phi.$$

由式 (3-27) 得

$$J_m = \nabla \times P_m = \nabla \times (\mu_r - 1) H = (\mu_r - 1) \nabla \times H$$

因此, 磁化电流体密度为

$$J_m = \begin{cases} (\mu_r - 1) J e_z, & a < r < b \\ 0, & 0 < r < a, r > b \end{cases}$$

在边界处, 由式 (3-28) 得

$$J_m = P_m \times e_n = (\mu_r - 1) H \times e_n$$

故在 $r = a^+$ 处, 磁化电流面密度为

$$\vec{\alpha}_m J_m = (\mu_r - 1) H e_\phi \times (-e_r) |_{r=a} = 0$$

在 $r = b^-$ 处, 磁化电流面密度为

$$J_m = (\mu_r - 1) H e_\phi \times e_r |_{r=b} = -(\mu_r - 1) \frac{J(b^2 - a^2)}{2b} e_z$$

由上式可知, 导波系统中 TEM 波的传播特性与无界空间中的均匀平面波的传播特性相同。另一方面, 将式 (8-1) 中的电场表达式代入波动方程 (6-2), 即 $\nabla^2 E + k^2 E = 0$, 得

$$\nabla_T^2 E + \frac{\partial^2 E}{\partial z^2} + k^2 E = \nabla_T^2 E + (\gamma^2 + k^2) E = \nabla_T^2 E + k_z^2 E = 0$$

其中, ∇_T 表示对横向坐标 x, y 求偏导。由于对于 TEM 波 $k_z^2 = 0$, 因此上式为

$$\begin{cases} \nabla_T^2 E = 0 \\ \nabla_T^2 H = 0 \end{cases}$$

由以上两式可见, TEM 波在横截面上的电磁场分布满足拉普拉斯方程, 因此 TEM 波在横截面上的电磁场特性与静电场一样。所以, 其导行波的场可用二维静电场分析法求出。

根据 8.1 节的讨论, 设加在同轴线内外导体上的电压为 U_0 , 在传播 TEM 波的情况下其横截面上的电磁场分布与静电场相同。同轴线中的电磁场分布也可以由麦克斯韦方程导出, 或者直接由拉普拉斯方程求得。在圆柱坐标系下, 电位的解满足一维拉普拉斯方程, 即

$$\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial \varphi}{\partial r} \right) = 0$$

利用边界条件 $r = a, \varphi = U_0; r = b, \varphi = 0$, 即可求得上述方程的解为

$$\varphi = \frac{U_0}{\ln \frac{a}{b}} \ln \frac{r}{b}$$

因此, 考虑到沿 z 方向的传播因子 $e^{-j\omega z}$, 电场和磁场为

$$\begin{cases} E_r = -\nabla \varphi e^{-j\omega z} = \frac{U_0}{r \ln \frac{a}{b}} e^{-j\omega z} e_r = \frac{E_0}{r} e^{-j\omega z} e_r \\ H_\phi = \frac{E_0}{\eta r} e^{-j\omega z} e_\phi \end{cases}$$

$$\begin{cases} U(z) = \int_a^b E_r dr = E_0 \ln \frac{b}{a} e^{-j\omega z} \\ I(z) = \oint_H H_\phi dl = \int_0^{2\pi} r H_\phi d\phi = \frac{2\pi E_0}{\eta} e^{-j\omega z} \end{cases}$$

静态磁场的安培环路定理

例 2.6 电容率为 ϵ 的介质球置于均匀外电场 E_0 中, 求电势。

解 介质球在外电场中极化, 在它表面上产生束缚电荷。这些束缚电荷激发的电场叠加在原外电场 E_0 上, 得总电场 E 。束缚电荷分布和总电场 E 互相制约, 边界条件正确地反映这种制约关系。

设球半径为 R_0 , 球外为真空 (图 2-5)。此问题具有轴对称性, 对称轴为通过球心沿外电场 E_0 方向的轴线, 取此轴线为极轴。

介质球的存在使空间分为两均匀区域——球外区域和球内区域。两区域内部都没有自由电荷, 因此电势 φ 都满足拉普拉斯方程。以 φ_1 代表球外区域的电势, φ_2 代表球内的电势。由 (2.3.3) 式, 两区域的通解为

$$\varphi_1 = \sum_n \left(a_n R^n + \frac{b_n}{R^{n+1}} \right) P_n(\cos \theta) \quad (2.3.11)$$

$$\varphi_2 = \sum_n \left(c_n R^n + \frac{d_n}{R^{n+1}} \right) P_n(\cos \theta) \quad (2.3.12)$$

a_n, b_n, c_n 和 d_n 是待定常量。边界条件包括:

(1) 无穷远处, $E \rightarrow E_0$, 由第一节例 1 得

$$\varphi_1 \rightarrow -E_0 R \cos \theta = -E_0 R P_1(\cos \theta) \quad (2.3.13)$$

因而

$$a_1 = -E_0, \quad a_n = 0 \quad (n \neq 1) \quad (2.3.14)$$

(2) $R = 0$ 处, φ_2 应为有限值, 因此

$$d_n = 0 \quad (2.3.15)$$

(3) 在介质球面上 ($R = R_0$):

$$\varphi_1 = \varphi_2, \quad e_n \frac{\partial \varphi_1}{\partial R} = e_n \frac{\partial \varphi_2}{\partial R} \quad (2.3.16)$$

2.3.11) 式和 (2.3.12) 式代入得

$$-E_0 R_0 P_1(\cos \theta) + \sum_n \frac{b_n}{R_0^{n+1}} P_n(\cos \theta) = \sum_n c_n R_0^n P_n(\cos \theta) \quad (2.3.17)$$

$$-E_0 P_1(\cos \theta) - \sum_n \frac{(n+1)b_n}{R_0^{n+2}} P_n(\cos \theta)$$

$$= \frac{\epsilon}{\epsilon_0} \sum_n n c_n R_0^{n-1} P_n(\cos \theta)$$

$$-E_0 R_0 + \frac{b_1}{R_0^2} = c_1 R_0$$

$$-E_0 - \frac{2b_1}{R_0^3} = \frac{\epsilon}{\epsilon_0} c_1 \quad (2.3.18)$$

由 (2.3.18) 式解出

$$b_1 = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} E_0 R_0^3, \quad c_1 = \frac{\epsilon - 3\epsilon_0}{\epsilon + 2\epsilon_0} E_0 \quad (2.3.19)$$

比较 (2.3.17) 式其他 P_n 项的系数可解出

$$b_n = c_n = 0, \quad n \neq 1 \quad (2.3.20)$$

所有常量已经定出, 因此本问题的解为

$$\begin{aligned}\varphi_1 &= -E_0 R \cos \theta + \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \frac{E_0 R_0^3 \cos \theta}{R^2} \\ \varphi_2 &= -\frac{3\epsilon_0}{\epsilon + 2\epsilon_0} E_0 R \cos \theta\end{aligned} \quad (2.3.21)$$

现在讨论此解的物理意义。由 (2.3.21) 式, 球内电场为 $\frac{3\epsilon_0}{\epsilon + 2\epsilon_0} E_0$, 因为 $\frac{3\epsilon_0}{\epsilon + 2\epsilon_0}$ 总小于 1, 所以在球内电场比原外电场 E_0 为弱, 这是由于介质球极化后在右半球面上产生正束缚电荷, 在左半球面上产生负束缚电荷, 故而在球内束缚电荷激发的场与原外场反向, 使总电场减弱。在球内总电场作用下, 介质的极化强度为

$$P = \chi' \epsilon_0 E = (\epsilon - \epsilon_0) E = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} 3\epsilon_0 E_0 \quad (2.3.22)$$

介质球的总电偶极矩为

$$p = \frac{4\pi}{3} R_0^3 P = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} 4\pi \epsilon_0 R_0^3 E_0 \quad (2.3.23)$$

(2.3.21) 式 φ_1 中的第二项正是这个电偶极矩所产生的电势

$$\frac{1}{4\pi \epsilon_0} \frac{p \cdot R}{R^3} = \frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \frac{E_0 R_0^3}{R^2} \cos \theta \quad (2.3.24)$$

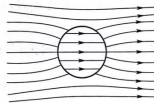


图 2-5